

Welcome to Course 2: Advanced Learning Algorithms

This course covers two of the most powerful and widely used machine learning algorithms: **Neural Networks** (also called Deep Learning) and **Decision Trees**.

Course Goals

1. **Learn Advanced Algorithms:** Implement and understand neural networks and decision trees.
 2. **Gain Practical Advice:** Learn how to make effective, systematic decisions when building real-world machine learning systems (e.g., deciding between collecting more data vs. building a bigger model) to avoid common pitfalls and wasted time.
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Course Structure (4 Weeks)

- **Week 1: Neural Networks (Inference)**
 - Understand the fundamentals of how neural networks work.
 - Learn how to use a pre-trained neural network to make predictions (a process called **inference**).
- **Week 2: Neural Networks (Training)**
 - Learn how to train your own neural network from scratch using a labeled dataset (X , Y).
- **Week 3: Practical Advice for Building ML Systems**
 - Focus on the best practices and strategic tips for building efficient and effective ML applications.
- **Week 4: Decision Trees**
 - Learn another highly powerful and widely used algorithm for classification and regression tasks.